

DESCRIPTION OF DESIGN DATA FOR FIBROUS COMPOSITES

Part: 1.4 – COMPOSITE MATERIALS
Chapter: 1411
Title: DESCRIPTION OF DESIGN DATA FOR FIBROUS COMPOSITES
Revision: 1.0
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1 INTRODUCTION

This chapter lists and describes the physical and mechanical properties required for stress analyses with fibrous composites (fibre reinforced polymers).

The methods applicable to calculate their design values from tests results data are presented in chapters 1480 “DERIVATION OF DESIGN DATA IN COMPOSITES” and 1481 “DERIVATION OF DESIGN DATA FOR FIBROUS COMPOSITES”.

Details on tests methods, on statistical methodologies, conventions and policies for data processing, are out of the scope of this document. See abovementioned chapters and the list of references for further information about these topics.

2 DEFINITIONS OF MATERIALS DESIGN DATA

2.1 IDENTIFICATION AND DESCRIPTION

Short (informal) Name	Eg. IMA/M21-UD
Provider and Commercial Name	
Material Designation(s) and Specification(s)	AIMS number, “grade”, ...Full designation of material in Airbus design documentation and existing equivalent designations
Description	Short description, including type of constituents, reinforcement style, manufacturing key aspects, ...
Version	Version of design values set

These data shall be provided by, or agreed with, Airbus M&P function (with the possible exception of the short informal name).

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2.2 PHYSICAL PROPERTIES

Symbol	Units	Description
CPT	mm	Cured Ply Thickness, i.e. representative nominal thickness of the cured ply
ρ	kg/m ³	Density
T _{g-on}	°C	Glass transition temperature (onset) of the material in moisture saturated condition
MOL	°C	Maximum Operational Limit, or maximum continuous service temperature without material irreversible degradation

All these data shall be provided by Airbus M&P function (with no further treatment at Stress function).

2.3 THERMAL EXPANSION AND PLY STIFFNESS

Coefficients of Thermal Expansion

Symbol	Units	Description
CTE _i α_i	1/°C	Coefficient of thermal expansion of material in a direction <i>i</i> (1 / fibre / warp, 2 / resin / weft, 3 / through-thickness) within a temperatures interval

The property in direction 3 (through-thickness) is normally not necessary for stress calculations. Mean/averaged values (*) shall be presented within temperature ranges of interest (see example table below) and for the whole service range.

Direction	Coefficients of Thermal Expansion (1/°C)			
	-55 to 20°C	20 to 70°C	70 to 120°C	All conditions
α_1				
α_2				
α_3				

Table 1. Example of presentation of CTE's

(*) However, excessive dispersion of results data within samples/batches shall have to reported and notified to the Stress function.

Elastic moduli

Symbol	Units	Description
E ₁	MPa	Elastic modulus in fibre/warp material direction
E ₂		Elastic modulus in resin/weft material direction
G ₁₂		In-plane Shear modulus
ν_{12}	-	Poisson's ratio
K _{BM}	-	B-basis correction factor for elastic moduli (E ₁ , E ₂ and G ₁₂) in buckling calculations

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Mean values of the elastic moduli shall be presented for the set of prescribed environmental conditions of interest, and/or a set of averaged values applicable across the whole temperature service range.

<i>Elastic Moduli – Mean values per condition</i>						
Property	Units	-55°C/Dry	RT/Dry	70°C/Wet	120°C/Wet	All conditions
E_1	MPa					
E_2						
G_{12}						
ν_{12}	-					

Table 1. *Example of presentation of elastic moduli*

- ✓ It is expected that the environmental knock down factor for fibre-driven moduli (E_1 but also E_2 in bi-axial fabrics) is 1.0, at least for temperatures not higher than the MOL. Minor differences shall be normally condensed into an unique value.
- ✓ It is also expected that in bi-axial fabrics E_1 and E_2 are equal. Test results shall be normally averaged into one unique value also.

Buckling B-basis correction factor K_{BM} is calculated dividing the B-basis compression modulus (in laminates with a relevant amount of fibre in load direction) by the nominal design elastic modulus.

- ✓ This factor shall be unique for property, i.e. tests data for different environmental conditions shall be pooled to get the K_{BM} . In case of bi-axial fabrics ($E_1 = E_2$), tests data in warp and weft directions shall be pooled also to get its value.

Some stress and strength analysis may need the elastic moduli of the material in through-thickness directions and planes.

Symbol	Units	Description
E_3	MPa	Elastic modulus in material through-thickness direction
G_{13}		Shear modulus of the material in plane 1-3
G_{23}		Shear modulus of the material in plane 2-3
ν_{13}	-	Poisson's ratio 13, contraction in dir-3 due stretching in dir-1
ν_{23}	-	Poisson's ratio 23, contraction in dir-3 due stretching in dir-2

Typically these properties are not directly obtained from dedicated tests, and it is sufficient to derive them indirectly from other properties of the material.

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2.4 PLY AND LAMINATE STRENGTH

Plain Strength

Symbol	Units	Description
F_{1t}	MPa	Tension ultimate stress of the ply in fibre/warp direction (also referred as X_t in literature)
F_{1c}		Compression ultimate stress of the ply in fibre/warp direction (also referred as X_c in literature)
F_{2t}		Tension ultimate stress of the ply in resin/weft direction (also referred as Y_t in literature)
F_{2c}		Compression ultimate stress of the ply in resin/weft direction (also referred as Y_c in literature)
F_{12}		In-plane shear ultimate stress of the FRP ply (also referred as S in literature)
ϵ_{t-all}	$\mu\epsilon$	Allowable tension strain of the material in fibre direction
ϵ_{c-all}		Allowable compression strain of the material in fibre direction
K_B	-	B-basis correction factor for strengths (to apply to mean values) in order to get a safe design allowable value

Strengths of the ply F_{1t} , F_{1c} , F_{2t} , F_{2c} , F_{12} shall be obtained using the same tests methods (and in many cases the same coupons) used for obtaining the elastic moduli.

- ✓ The B-basis correction factors, for each property, shall be obtained after pooling tests data across the different environment conditions to get one value per property
- ✓ It is expected that in bi-axial fabrics axial strengths in warp and weft directions are equal. Test results shall be normally minored into one value for the tension strength ($F_{1t} = F_{2t}$) and another for the compression strength ($F_{1c} = F_{2c}$).

Allowable strains of the material shall be obtained from the ultimate stresses of the ply in fibre direction (F_{1t} , F_{1c}), and from the plain strengths of multiangular laminates (at least data for the quasi-isotropic laminate should be available from qualification tests), as the minimum ultimate strain calculated from these stresses and the applicable elastic moduli.

$$\epsilon_{t-all} = \min [F_{1t}/E_1, (PTS/E_x)_i]$$

$$\epsilon_{c-all} = \min [F_{1c}/E_1, (PCS/E_x)_i]$$

Property	Units	Ply strengths – Mean values per condition				K_B
		-55°C/Dry	RT/Dry	70°C/Wet	120°C/Wet	
F_{1t}	MPa					
F_{1c}						
F_{2t}						
F_{2c}						
F_{12}						
ϵ_{t-all}	$\mu\epsilon$					
ϵ_{c-all}						

Table 2. Example of presentation of ply strengths

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Property	Units	Ply strengths Mean at RT/Dry	K_B	Environmental Knock Down Factors		
				-55°C/Dry	70°C/Wet	120°C/Wet
F_{1t}	MPa					
F_{1c}						
F_{2t}						
F_{2c}						
F_{12}						
ϵ_{t-all}	$\mu\epsilon$					
ϵ_{c-all}						

Table 3. Alternative presentation of ply strengths using reference RT/Dry data and EKDF's

Some strength analysis (delamination checks) may need the ultimate strengths of the material in through-thickness directions and planes.

Symbol	Units	Description
F_{13}	MPa	Interlaminar shear ultimate stress of the ply in through thickness shear plane 1-3 (also referred as ILSS in literature)
F_{23}		Interlaminar shear ultimate stress of the ply in through thickness shear plane 2-3
F_{3t}		Interlaminar through-thickness tension ultimate stress (also referred as unfolding strength)
F_{3c}		Interlaminar through-thickness compression ultimate stress

Interlaminar shear strength F_{13} and unfolding strength F_{3t} can be obtained from tests on uni-directional laminates. F_{23} and F_{3c} are indirectly derived from other properties (from the ILSS strength in plane 1-3 and for the compression strength in dir. 2 in case of UD-tape materials).

Effect of environmental conditions and B-basis correction is treated in a similar manner as shown above for in-plane ply strengths.

Fracture toughness

Design Values for critical energy release rates, or fracture toughness, in fracture modes I (opening) and II (sliding / interlaminar shear) shall be provided for advanced numerical simulations.

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Bolted Joints

The parameters needed for feeding the analysis method for determining the strength of mechanically fastened joints in composite laminates, in addition to in-plane moduli and strengths, shall be:

Symbol	Units	Description
FHT	MPa	Filled Hole Tension strength of the quasi-isotropic laminate
FHC		Filled Hole Compression strength of the quasi-isotropic laminate
F _{BR}		Bearing strength of the quasi-isotropic laminate (failure at 6% hole deformation)
F _{PO}		Pull-out strength, defined as the allowable through-thickness mean shear stress in the laminate, across a cylindrical section of the same diameter as the rivet-head or nut, caused by a tension force in the fastener: $F_{PO} = P_{PO} / (\pi \cdot D_h \cdot t)$

Notice all those properties (with the exception of the pull-out strength) are obtained from test coupons made of one plate laminate (double-shear, no eccentricity effects) and with protruding rivets/bolts installed.

Residual Strength After Impact

The method for determining the residual strength of composite laminates, including a low-velocity transverse impact damage, is discussed in chapter 1421 "MODEL FOR RESIDUAL STRENGTH AFTER IMPACT IN COMPOSITE LAMINATES":

- ✓ The model currently under discussion proposes a fixed value of TAI as allowable strain in tension (not dependent of laminate thickness, stacking or dent depth visibility threshold).
- ✓ In compression, the model for CAI proposes and approximation function of laminate stacking and thickness, cut-off energy and dent depth visibility threshold.

2.5 FATIGUE AND DAMAGE TOLERANCE (NO-GROWTH) OF LAMINATES

Reserved for future use.

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3 DATASHEET TEMPLATE FOR A FRP MATERIAL

MATERIAL IDENTIFICATION

Short informal name	
Commercial name	
Specifications	
Description	
Version	

REFERENCES

- ...

PHYSICAL PROPERTIES

Cured Ply Thickness (CPT)	mm
Density	kg/m ³
Tg-onset (wet)	°C
Max. Operational Limit (MOL)	°C

Coefficients of Thermal Expansion (1/°C)	Range	-55..20°C	20..70°C	70..120°C
	α_1			
	α_2			
	α_3			

ELASTIC MODULI

Property	E_1	E_2	G_{12}	ν_{12}	K_B
Units	MPa	MPa	MPa	-	-
Values					

Remarks:

- The K_B is the buckling correction factor for moduli to apply to E_1 , E_2 and G_{12} in buckling stability analyses

Property	E_3	G_{13}	G_{23}	ν_{13}	ν_{23}
Units	MPa	MPa	MPa	-	-
Values					

PLAIN STRENGTHS

Ultimate stresses of the ply (from UD-laminates testing).

Property	Mean Values (MPa)				B-basis K_B	E-KDF			B-basis Values (MPa)			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
F_{1t}												
F_{1c}												
F_{2t}												
F_{2c}												
F_{12}												

Design values for checking the laminate strength using the modified Maximum Strain Criterion.

Property	Mean Values ($\mu\epsilon$)				B-basis K_B	E-KDF			B-basis Values ($\mu\epsilon$)			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
ϵ_{tail}												
ϵ_{call}												

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INTERLAMINAR PLY STRENGTHS

Property	Mean Values				B-basis K _B	E-KDF			B-basis Values			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
F ₁₃ (MPa)												
F _{3t} (MPa)												
G _{IC} (N/mm)												
G _{IIC} (N/mm)												

OPEN HOLE AND BOLTED JOINTS

Reference material strengths in open-hole, filled-hole and bearing for the quasi-isotropic laminate.

Property	Mean Values (MPa)				B-basis K _B	E-KDF			B-basis Values (MPa)			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
OHT _{QI}												
OHC _{QI}												
FHT _{QI}												
FHC _{QI}												
F _{BR-QI}												
F _{PO}												

COMPOSITE DAMAGE TOLERANCE

Design values for residual strengths after BVID impact in tension and compression (as well as approximation for dent depth to impact energy).

Property	B-basis Value for all conditions
TAI (μ€)	

$$E/t^2 = A \cdot (1 - e^{-B \cdot d_0/t})$$

Approximation model type: SMM chapter 1421: CDTM-T05

$$CAI = K_B \cdot EKDF \cdot \{ CAI_{\infty-QI} \cdot (3 / K_{TOH})^{C_x} + A / e^{B_e \cdot (E/t^2)} \}$$

Property	Design Value
A (J/mm ²)	
B	

Property	Mean Value RT/Dry	B-basis K _B	E-KDF		
			70°C/W	90°C/W	120°C/W
CAI _{∞-QI} (μ€)					
A (μ€)					
B _e					
C _x					

...where:

- E is the impact energy, not higher than the cut-off identified during damage threats analysis (for many parts of the airframe, 35J as cut-off value shall be acceptable)
- t the laminate thickness
- d₀ the dent depth after impact (typ. 2.0-2.5mm for BVID detectability under a GVI inspection)
- K_{TOH} is the stress Concentration factor for an infinite plate with a circular hole in by-pass load, calculated as from laminate apparent moduli (along X direction) as:

$$K_{TOH} = 1 + \{ 2 \cdot [(E_x/E_y)^{0.5} - \nu_{xy}] + E_{xy}/G_{xy} \}^{0.5}$$

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4 APPENDICES

OTHER ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
1	-	Ply direction parallel to fibres in unidirectional tape materials, and in warp direction in fabrics, as subscript
2	-	Ply direction normal to fibres in unidirectional tape materials, and in weft direction in fabrics, as subscript
3	-	Ply through-thickness direction, as subscript
CFC	-	Carbon Fibre Composites
CFRP	-	Carbon Fibre Reinforced Polymers
D _h	mm	External diameter of a rivet-head or nut in a fastener joint
EKDF / K _{EKDF}	-	Environmental Knock Down Factor to account for degradation of a mechanical property with material saturated in an ambient (temperature and humidity) different to the reference conditions of 20°C (Room Temperature or RT) and dry material
E _x	MPa	Apparent elastic modulus of a composite laminate, obtained from the nominal values of the elastic moduli of the plies and the stacking sequence, using Classic Laminate Theory
F	MPa	Ultimate/allowable stress of the material (typ. followed by subscript indicating the kind of load & failure)
FRP	-	Fibre Reinforced Polymers
GFRP	-	Glass Fibre Reinforced Polymers
K _B	-	B-basis Knock Down Factor for the minoring to the mean value of a mechanical property in order to get a safe design allowable value accounting for natural dispersion
OoA	-	Out of autoclave curing materials
P	N	Ultimate/allowable force
PTS	MPa	Plain tensile strength, or ultimate tensile stress, of a multiangular laminate
PCS	MPa	Plain compression strength, or ultimate compression stress, of a multiangular laminate
PO	-	Pull-Out, as subscript
QI	-	Quasi-Isotropic laminate
t	mm	Thickness of the laminate
TBC	-	To Be Confirmed
UD	-	Uni-Directional laminate

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REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[SAE] CMH-17	2012	COMPOSITE MATERIALS HANDBOOK
Ref. 2	[Airbus] PP0300181	Is. 2	Composite statistical method to determine B-basis value from small data sets"
Ref. 3	[Airbus] RP0412191	Is. 1	Normalization rules for composite properties
Ref. 4	TAE-T-MM-170009	Is. C	GUIDELINES FOR RESULTS ANALYSIS OF COUPONS TESTS IN COMPOSITE MATERIALS
Ref. 5	TEA-T-NT-180036	Is. A	LAMINATES STRENGTH. DISCUSSION ON TESTS RESULTS FOR IMA/M21E: PLAIN STRENGTH AND RESIDUAL STRENGTH AFTER IMPACT.
Ref. 6	TEA-T-MM-190002	Is. A	GUIDELINES FOR PRESENTATION OF RESULTS OF COUPONS TESTS IN COMPOSITE MATERIALS

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 4. Software programs referenced in this chapter

DESCRIPTION OF DESIGN DATA FOR FIBROUS COMPOSITES

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub- chapters affected
1.0 29/07/2020	G. Baños	First issue	All pages

Table 5. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Methods (TEPAP-TL4)	G. Baños 28/07/2020	Florian Glock 29/07/2020

Table 6. *Approval signatures table for last revision of this chapter*